Team Vyflare Synthetic Vial Image Generation

CSI Digital Twin Engineering Technical Report - Part Two

NVIDIA Rockwell Physical AI for the Factory Floor Hackathon

600	12	50	100%
verified vial images	display-color settings	authored frames per color	readable 640 x 480 PNGs

Prepared for	Team Vyflare and Amy's vial-image research
Platform	NVIDIA Brev, Isaac Sim 6.0.1, CSI fill station, FillCam
Report period	Part Two work completed 26-27 October 2026
Repository	brev-research-dataset branch, commit 4703b852
Data status	Synthetic experiment with held-out real-data evaluation required

Outcome: Team Vyflare expanded the verified synthetic corpus with a new standalone 600-image dataset, preserved the original digital twin, published both extracted files and a Git LFS archive, and documented the dataset for reproducible model research.

Interpretation boundary: synthetic colors are display settings only. They do not encode a recipe, chemical, mixture ratio, material identity, quality state, or defect. Authored animation frames are not calibrated fill percentages or measured liquid volumes.

1 Executive Summary

Part Two converted the earlier proof-of-concept workflow into a larger, balanced, developer-ready dataset. The new run crossed 12 synthetic display-color settings with 50 distinct authored fill-station frames, yielding 600 images. The generation process used the existing FillCam view at 640 x 480 pixels and 32 path-tracing subframes. Camera, lighting, and vial pose remained fixed because the verified renderer exposes only authored frame and liquid diffuse-color controls.

A 120-image pilot was rendered first and visually inspected through a 12-color contact sheet. After the pilot passed count, readability, and visual checks, the remaining 480 cases were rendered. A finalizer validated every image, copied verified outputs into per-color folders, produced JSONL and CSV metadata, and recorded source-integrity results. The original USD and USDC assets were not overwritten.

Progression from proof of concept to research handoff

Milestone	Design	Images	Purpose
Part One verified corpus	Baseline, temporal, and color sweeps	240	Prove non-destructive generation and traceability
Part Two pilot	10 frames x 12 colors	120	Check appearance and estimate runtime before scaling
Part Two remaining run	40 frames x 12 colors	480	Complete balanced 50-frame coverage
Part Two deliverable	50 frames x 12 colors	600	Standalone synthetic research dataset
Known project total	Part One outputs retained plus Part Two	840	Full-quality images across separate output sets

The 600-image dataset should be treated as controlled synthetic augmentation. Model selection and final claims must use held-out real vials or entire held-out real acquisition sessions.

Part Two deliverables

- Twelve color folders with exactly 50 verified images each.
- One JSONL record and one CSV row for every image.
- A verification summary, dataset README, and representative contact sheet.
- A Git LFS archive for developer download plus extracted files on the research branch.
- A reproducible two-stage render plan that can be rerun on Brev.

2 Dataset Design and Visual Coverage

The dataset uses a complete 12-by-50 grid: every display color appears at the same 50 authored frames. This balanced layout supports controlled comparisons because color coverage and frame coverage are identical across folders. Frame values span 30 through 686 and were selected by evenly sampling the authored sequence then rounding to unique integer frames.

Color	sRGB	Images	Color	sRGB	Images
Navy	#2446A8	50	Yellow	#F4D03F	50
Blue	#2078D9	50	Amber	#F2A51A	50
Cyan	#20C7D9	50	Orange	#E76F2E	50
Teal	#159A9C	50	Red	#D94A4A	50
Green	#35B779	50	Magenta	#C64BD8	50
Lime	#8BC34A	50	Violet	#7656C9	50

All hexadecimal values above are intended render parameters. They have no recipe, chemistry, ratio, quality, or defect meaning.

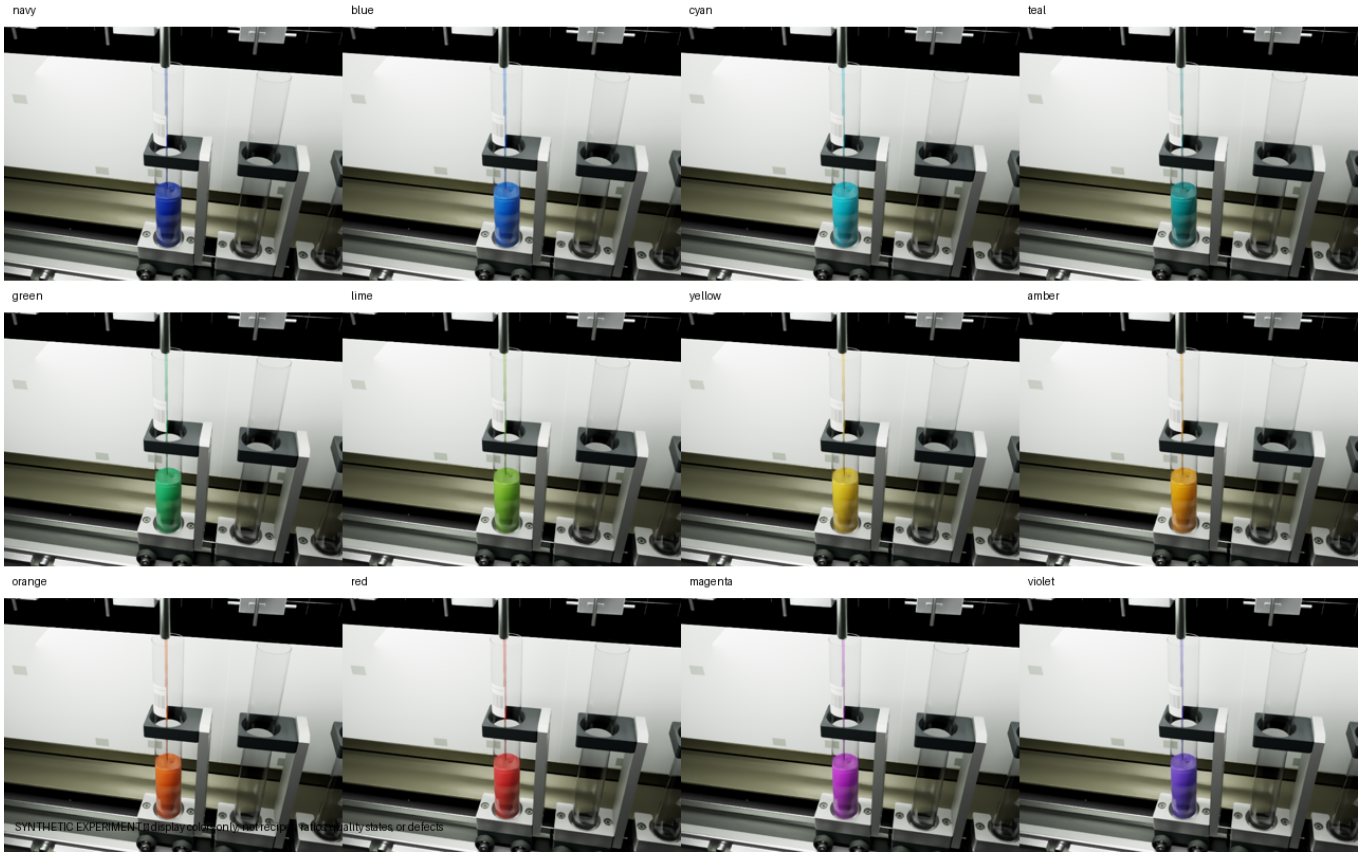


Figure 1. Pilot contact sheet at authored frame 298. Geometry and viewpoint remain fixed while the liquid display-color parameter changes across all 12 settings.

Authored frame coverage

30, 43, 57, 70, 84, 97, 110, 124, 137, 150, 164, 177, 191, 204, 217, 231, 244, 258, 271, 284, 298, 311, 325, 338, 351, 365, 378, 391, 405, 418, 432, 445, 458, 472, 485, 499, 512, 525, 539, 552, 566, 579, 592, 606, 619, 632, 646, 659, 673, 686

3 Engineering Workflow on NVIDIA Brev

The workflow was intentionally staged to limit risk and make failures recoverable. Raw renders were written to isolated staging directories. Only after count, decode, dimension, and source-integrity checks passed were images copied into the final color folders.



Figure 2. Staged dataset-generation and promotion workflow.

Measured execution

Run	Cases	Measured time	Result
Pilot	120	129 seconds	120 readable PNGs; 10 per color; contact sheet passed
Remaining	480	452 seconds	480 readable PNGs; completed 50 frames per color
Combined	600	581 seconds	Approximately 9.7 minutes of measured render runtime

Non-destructive controls

Control	Implementation	Evidence
Temporary material edits	Author inputs:diffuseColor in an anonymous USD session layer	usd_edit_layer recorded as anonymous_session_layer
No source save	Renderer never saves source USD or session layers	source_files_saved = false in both runs
Before/after integrity	Compare protected source state before and after each run	source_before equals source_after
Isolated outputs	Write pilot and remaining runs under _staging	Final folders populated only after validation
No invented variation	Keep camera, lighting, and vial pose fixed	Metadata records fixed_existing_scene_not_varied

Only two verified variables changed: authored animation frame and liquid diffuse display color. The workflow did not claim camera, lighting, vial-pose, physical-volume, or process randomization.

4 Metadata, Quality Assurance, and Traceability

Each image is represented in dataset_manifest.jsonl and dataset_manifest.csv. The schema records the render controls, fixed-scene assumptions, file identity, generation time, and interpretation warnings needed to prevent accidental conversion of synthetic settings into physical labels.

Metadata group	Representative fields	Why it matters
Identity	image, case_id, source_run, sha256	Connects every image to one reproducible render case
Animation	frame, fill_level_setting, fill_level_units	Preserves authored frame while denying measured-volume interpretation
Color	synthetic_color_name, intended_hex_color_srgb, liquid_color_control	Documents the exact display parameter without assigning chemistry
Imaging	FillCam prim, 640 x 480 resolution, subframes	Records camera and rendering context
Fixed conditions	lighting_settings, vial_position_or_orientation, random_seed	Makes absent randomization explicit
Provenance	source_stage, source_script, generation_datetime_utc	Supports audit and regeneration
Validity	valid_image, source_files_saved, usd_edit_layer	Captures image checks and source-protection state

Final verification results

Verification check	Observed result
Total images	600
Per-color balance	50 images in each of 12 folders
Frame uniqueness	50 distinct authored frames per color
Image decoding	600 of 600 readable
Image dimensions	600 of 600 at 640 x 480
Metadata rows	600 JSONL records; 600 CSV data rows
Pilot visual review	Vial visible and all 12 display-color settings distinguishable
Source integrity	Protected source state unchanged for pilot and remaining runs

Archive SHA-256: 70E191ABC592266674805B60F27F59EDD4C114962295B54F483796E3AA34647D

5 GitHub Delivery and Developer Handoff

The original ZIP is 137,743,764 bytes, which exceeds GitHub's normal 100 MiB Git object limit. Uploading it as a normal repository file caused commit failures in the browser-based Remote Repositories workflow. The solution was to clone the private repository locally, configure Git LFS for the ZIP only, commit the extracted dataset as regular files, and push the result to the existing research branch.

Item	Published location	Delivery method
Extracted images	data output//*.png	Regular Git files; 50 per color
Metadata and QA	data output/*.jsonl, *.csv, *.json, README.md	Regular Git files
Representative visual	data output/pilot_contact_sheet.png	Regular Git file
Developer archive	data output.zip	Git LFS object, 137,743,764 bytes
Repository guidance	Root README.md and .gitattributes	Documents LFS pull and research limitations

Published revision

Repository	ALL-Applied-AI-Network/msoe-ai-club-rockwell-automation-nvidia-hackath-vyflare-9077472263
Branch	brev-research-dataset
Commit	4703b8523fb14436561432cfab8a8267ccd55ec7
Commit message	Add 600-image synthetic vial research dataset
Files changed	608

Developer checkout

```
git clone --branch brev-research-dataset --single-branch <repository-url>
cd msoe-ai-club-rockwell-automation-nvidia-hackath-vyflare-9077472263
git lfs install
git lfs pull
```

Developers who need only the extracted images can use data output/ directly. Developers who want the original handoff archive should install Git LFS before pulling the branch.

The ZIP and extracted folder contain the same dataset. Git LFS prevents the oversized archive from becoming an invalid ordinary Git blob while preserving a single-file download for collaborators.

6 How This Supports Amy's Vial-Image Research

The dataset is most useful as a controlled synthetic supplement. It increases coverage of visual liquid appearance across the authored fill-station sequence while keeping camera and scene geometry consistent. This makes it suitable for pretraining, data-pipeline validation, broad appearance sensitivity studies, and augmentation experiments.

Experiment	Training data	Selection data	Final evaluation
Real-only baseline	Real training sessions	Real validation sessions	Untouched real test sessions
Synthetic pretraining	600 synthetic images, then real fine-tuning	Real validation sessions	Same untouched real test sessions
Mixed training	Real data plus a controlled synthetic fraction	Real validation sessions	Same untouched real test sessions

Recommended evaluation principles

- Split real data by vial or acquisition session so near-duplicate frames cannot leak across train and test.
- Use real validation data for hyperparameter selection and early stopping.
- Report the real-only baseline beside synthetic-pretraining and mixed-training results.
- Track performance by real session, lighting condition, vial state, and camera condition when metadata allows.
- Treat an improvement on held-out real data as the success criterion, not a high synthetic validation score.

Known domain-gap risks

Risk	Current Part Two state	Mitigation
Fixed viewpoint	Only FillCam was used	Compare against real camera variation before adding pose randomization
Fixed lighting	Existing scene lighting was retained	Measure real illumination and reflection ranges
No sensor model	No noise, blur, focus, or compression variation	Estimate these distributions from real data
Frame-label ambiguity	Frame is not calibrated volume	Use frame only as a synthetic control variable
Color-label ambiguity	Color is not a recipe or defect	Keep display-color labels separate from real process labels

7 Reproduction, Handoff Paths, and Next Steps

Verified Brev reproduction sequence

```
cd /workspace/csi_digital_twin_v1
python3 tools/generate_12color_manifests_v5.py
/isaac-sim/python.sh tools/render_synthetic_experiments.py --config experiments/synthetic_12color_pilot_v5.json
--out-dir '/workspace/data output/_staging/pilot_raw' --subframes 32
/isaac-sim/python.sh tools/render_synthetic_experiments.py --config
experiments/synthetic_12color_remaining_v5.json --out-dir '/workspace/data output/_staging/remaining_raw'
--subframes 32
/isaac-sim/python.sh tools/finalize_12color_dataset_v5.py
```

Item	Path or identifier
Brev project	/workspace/csi_digital_twin_v1
Brev final dataset	/workspace/data output
Pilot configuration	experiments/synthetic_12color_pilot_v5.json
Remaining configuration	experiments/synthetic_12color_remaining_v5.json
Renderer	tools/render_synthetic_experiments.py
Manifest generator	tools/generate_12color_manifests_v5.py
Finalizer	tools/finalize_12color_dataset_v5.py
GitHub branch	brev-research-dataset

Recommended next steps

- Create an inventory of Amy's real vial images by session, vial, camera setup, image dimensions, and trusted label availability.
- Measure real distributions for illumination, glare, focus, blur, sensor noise, occlusion, camera pose, and vial placement.
- Define a held-out real-session benchmark before using the synthetic images for training decisions.
- Run real-only, synthetic-pretraining, and mixed-training comparisons with identical real validation and test sets.
- Extend the generator only for domain gaps supported by real evidence, recording every new variable in metadata.
- Keep dataset releases versioned and preserve source-integrity checks for every future render sweep.

Part Two is complete: the balanced 600-image dataset is generated, verified, documented, and available to the team as both extracted files and a Git LFS archive. The next milestone is evidence-based comparison against held-out real vial data.